

Executive Summary

Meeting of ITU-T SG17 'Security', Geneva, 21 February – 3 March 2023

Hot topics

- Intelligent transport system security
- 5G security
- IoT security
- Cloud security
- Cybersecurity
- Quantum based security
- Simulation for security

1. Meeting outputs (meeting statistics see Annex E below)

- Output standards (20, see Annex A):
 - TAP approval (9): Details are in Annex A a).
 - TAP (re-)determined (6): 5 new Recommendations and 1 Technical Corrigendum. Details are in Annex A b).
 - AAP consented (4): 4 new Recommendations for AAP Last Call. Details are in Annex A c).
 - Agreed (1): 1 new Technical Report. Details are in Annex A d).
- New work items (23, see Annex C).
- Discontinued work items (1, see Annex D).
- [ITU Workshop on "DLT security, identity management and privacy"](#) was held successfully on 20 February 2023
- Replacement of the SG17 vice-chairman from Algeria
- Replacement of WP3/17 vice-chairman by Afnan Alromi (Saudi Arabia)
- Survey on countering spam fraud launched
- SG17 sub-groups:
 - JCAs:
 - JCA-COP: remains dormant
 - [JCA-IdM](#): continued with revised ToR, and held its 31st meeting on Friday 24 February 2023
 - (JCA-DCC: meeting on Friday 24 February 2023)
 - [Correspondence Groups](#)
 - CG-SG17-wtsa24-prep: continued
 - CG-secapa (Correspondence Group on Security Capability and Architecture): continued
 - CG-FIDO (Correspondence Group on FIDO standards): terminated
 - Advisory group on X.509: continued

2. Next SG17 meetings

- **WP3/17 plenary**, Fully virtual, 8 May 2023 to consent revised ITU-T X.1051 |ISO/IEC 27011
- **2nd ITU X.509 Day event**, Fully virtual, 9 May 2023

- **4th SG17 meeting**, Korea, Tuesday 29 August - Friday 8 September 2023 (9 working days, physical meeting with remote participation)
 - **ITU workshop on "Zero trust and identity management security"**, Monday 28 August 2023, 09:30-17:30
 - **Open and extended management team meeting**, Monday 28 August 2023, 19:00-21:00.
 - **32nd JCA-IdM meeting**, Friday 1 September 2023, 14:30-16:00
 - **24** Candidate texts for action, details see Annex B.
- (ITU-WHO workshop on DCC (TBD) on 11 September 2023 followed by 5th JCA-DCC meeting on 12 September 2023.)
- **5th SG17 meeting**: Feb/Mar 2024, Malaysia (tbc)

3. Interim RGMS

7 Questions plan to hold the following 9 RGMs before next SG17 meeting:

#	Q/17	Date	Place/Host	Subject/objective
1	2/17	18-19 May 2023	Seoul, Korea (Hosted by SCH University)	• To prepare texts for action in next SG17 meeting: X.5Gsec-ctrl and TR.cpn-col-sec. • To review all work items and identify future topics for Q2/17
2	3/17	13 April 2023	e-meeting	• X.1051rev2
3	3/17	13 -14 June 2023	e-meeting	X.sup-cdc (Clause 7: Context: CDC in the bigger context of operational security)
4	4/17	29 June 2023	e-meeting	• To address and discuss X.stie and X.taeii
5	10/17	6 June 2023	e-meeting	• progress all the work of Q10/17
6	11/17	20 - 24 March 2023	Wien, Austria	• Generic technologies (such as Directory, PKI, formal languages, object identifiers) to support secure applications. • Draft agenda is in TD923R1
7	13/17	14 - 15 June 2023	e-meeting	• prepare the final texts for X.itssec-5, X.1373rev • address all work items and future topics.
8	15/17	30 May - 1 June 2023	Singapore (Hosted by IMDA/NUS)	• QKD related WIs
9	15/17	7 June 2023	e-meeting	• X.icd-schemas

Annex A

Actions taken on Recommendations, and other texts at SG17 closing plenary on 3 March 2023

a) TAP Recommendations approved (WTSA-20 Resolution 1)

#	Q	Acronym	Title	New / Revised	Editor(s)	Location of text	Equivalent e.g., ISO/IEC	Start of work	Timing
1.	2/17	X.1815 (X.5Gsec-ecs)	Security guidelines and requirements for IMT-2020 edge computing services	New	Feng GAO, Jae Hoon NAH, Junjie XIA, Bo YU, Xiaojun ZHUANG	TD1015		2019-01	2022-09
2.	2/17	X.1816 (X.5Gsec-ssl)	Guidelines and requirements for classifying security capabilities in IMT-2020 network slice	New	Zhiyuan HU, Li SU, Ke WANG, Bo YANG	TD915		2020-09	2022-09
3.	8/17, (14/17)	X.1411 (X.BaaS-sec)	Guidelines on blockchain as a service (BaaS) security	New	Nan MENG, Kyeong Hee OH, Zixiang WANG, Xuan ZHA	R20		2019-09	2022-09
4.	8/17	X.1644 (X.sgdc)	Security guidelines for distributed cloud	New	Mark MCFADDEN, Ye TAO, Lei XU, Laifu WANG	R21		2019-01	2022-09
5.	13/17	X.1380 (X.edr-sec)	Security guidelines for cloud-based data recorders in automotive environments	New	Sang-Woo LEE, Seungwook PARK	R22		2018-03	2022-09

6.	13/17	X.1381 (X.eivn-sec)	Security guidelines for Ethernet-based In-Vehicle networks	New	Sang-Woo LEE, You-Sik LEE	TD917		2018-09	2022-09
7.	13/17	X.1382 (X.fstiscv)	Guidelines for sharing security threat information on connected vehicles	New	Min SHU, Wenlei WANG, Xiaochun YUN, Yunwei ZHAO	TD918		2018-09	2022-09
8.	13/17	X.1383 (X.srzd)	Security requirements for categorized data in vehicle-to-everything (V2X) communication	New	Takamasa ISOHARA, Nan MENG, Yaping SUN, Huirong TIAN	TD919		2018-09	2022-09
9.	14/17	X.1410 (X.sa-dsm)	Security architecture for data-sharing management based on the distributed ledger technology	New	Feng GAO, Zhiyuan HU, Min SHU, Yunwei ZHAO	TD900		2020-09	2022-09

b) TAP Recommendations determined (WTSA-20 Resolution 1)

#	Q	Acronym	Title	New / Revised	Editor(s)	Location of Text	Equivalent e.g., ISO/IEC	Start of work	Timing
1	2/17	X.1817 (X.5Gsec-message)	Security requirements for 5G message service	New	Hang DONG, Le YU, Hongyang ZHANG	TD939 (A.5 in TD934)		2021-04	2023-03

2	6/17	X.1333 Cor. 1	Corrigendum 1 to X.1333: Security guidelines for use of remote access tools in Internet- connected control systems	New	Gunhee Lee	TD958		2023- 02	2023- 03
3	6/17	X.1353 (X.ztd-iot)	Security methodology for zero- touch deployment in massive IoT based on blockchain	New	Xin KANG, Haiguang WANG, Weidong WANG	TD937		2020- 09	2022- 09
4	7/17	X.1471 (X.websec- 7)	Reference monitor for online analytics services	New	Jongyoul Park, Junjie Xia, Hyungjin Lim, Jah Hoon Nah	TD936		2014- 09	2023- 03
5	7/17	X.1771 (X.rdda)	Requirements for data de- identification assurance	New	Feng Gao, I Seok Kang, Soonseok Kim, Jihun Kim, Byunghoon Lee, Yunsik Park, Hyungjin Lim, Heung Youl Youn	TD1012		2019- 01	2023- 03
6	8/17	X.1645 (X.nssa-cc)	Requirements of network security situational awareness platform for cloud computing	New	Maofei CHEN, Huamin JIN, Zhaoji LIN, Laifu WANG, Yi ZHANG	TD965		2019- 09	2023- 03

c) AAP Recommendations consented (Recommendation ITU-T A.8)

#	Q ⁽¹⁾	Acronym	Title	New / Revised	Editor(s)	Location of Text	Equivalent e.g., ISO/IEC	Start of work	Timing
1	4/17	X.1219 (ex X.arc-ev)	Functional requirements for a secured process to evaluate technical vulnerabilities	New	Wei Li, Shan Xue, Chen Zhang	TD930		2019-08	2023-03
2	10/17	X.1278.2 (X.ctap21)	Client to authenticator protocol version 2.1	New	Abbie Barbir	TD886 (A.5 in TD887)	FIDO CTAP2.1	2022-05	2023-03
3	10/17	X.1277.2 (X.uaf12)	Universal authentication framework version 1.2	New	Abbie Barbir	TD888 (A.5 in TD889)	FIDO UAF1.2	2022-05	2023-03
4	14/17	X.1412 (X.srscm-dlt)	Security Requirements for Smart Contract Management based on the distributed ledger technology	New	Kepeng Li, Kyeong Hee Oh, Dong Bin Choi, Yang Wu, Min Shu	TD920		2020-09	2023-02

d) Non-normative texts (Technical Report, Supplement, Implementers' Guide, etc) agreed

#	Q	Acronym	Title	New / Revised	Editor(s)	Location of text	Equivalent e.g., ISO/IEC	Start of work	Timing
1.	6/17	TR.ibc-cd	Guidelines for identity based cryptosystems used for cross-domain secure communications	New	Fuwen Liu, Li Su, Junzhi Yan, Bo Yang	TD956		2021-04	2023-03

Annex B

Recommendations planned for action in SG17 Aug/Sep 2023 meeting

a) TAP Recommendations planned for TAP approval (WTSA-20 Resolution 1)

#	Q	Acronym	Title	New / Revised	Editor(s)	Location of Text	Equivalent e.g., ISO/IEC	Start of work	Timing
1.	7/17	X.1454 (X.sles)*	Security measures for location enabled smart office services	New	Junjie Xia, Feng Gao, Jae Hoon Nah, Hang Dong	TD985		2019-09	2022-09

b) TAP Recommendations planned for TAP determination (WTSA-20 Resolution 1)

#	Q ⁽¹⁾	Acronym	Title	New / Revised	Editor(s)	Location of Text	Equivalent e.g., ISO/IEC	Start of work	Timing
1	2/17	X.5Gsec-ctrl*	Security controls for operation and maintenance of 5G network systems	New	Ayumu KUBOTA, Koji NAKAO, Yutaka MIYAKE	TD1016		2022-09	2023-09
2	4/17	X.spmoh*	Security framework for storage protection against malware attacks on hosts	New	Jonghyun Woo, Bongchan Kim Heejun Shin Jonghyun Kim Sujung Park	TD929		2022-05	2023-09

3	4/17	X.sr-ctea*	Security requirements and countermeasures for targeted email attacks	New	Chunghan Kim Jonghyun Kim Sujung Park	TD924		2022-05	2023-09
4	4/17	X.stie*	Structured Threat Information Expression	New	Michael ROSA, Duncan SPARRELL	TD972	OASIS STIX Version 2.1	2022-09	2023-09
5	4/17	X.taeii*	Trusted Automated Exchange of Intelligence Informa	New	Michael ROSA, Duncan SPARRELL	TD971	OASIS TAXII Version 2.1	2022-09	2023-09
6	6/17	X.sc-iot*	Security Controls for Internet of Things (IoT) systems	New	Koji Nakao, Liu Lijun	TD1032		2018-09	2023-09
7	10/17	X.oob-sa*	Framework for out-of-band server authentication using mobile devices	New	Il Jin JUNG, Sujung PARK, Heejun SHIN, Jonghyun WOO	TD1044		2022-01	2023-09
8	13/17	X.itssec-5*	Security guidelines for vehicular edge computing	New	Sang-Woo Lee	TD981		2017-09	2023-09
9	13/17	X.1373rev*	Secure software update capability for intelligent transportation system communication devices	Revised	Koji Nakao, Sang-Woo Lee, Aram Cho, Seungwook Park	TD997		2018-08	2023-09

Notes:

(1) In case of joint Question activity, the lead Question is given without parentheses and other Questions are shown in parentheses; such entries are only shown in the table against the lead Question.

c) AAP Recommendations planned for AAP consent (Recommendation ITU-T A.8)

#	Q ⁽¹⁾	Acronym	Title	New / Revised	Editor(s)	Location of Text	Equivalent e.g., ISO/IEC	Start of work	Timing
1	7/17	X.1144rev	The revision of eXtensible Access Control Markup Language (XACML) 3.0	Rev	Jae Hoon NAH, Duncan Sparrell	TD1033		2020-03	2023-09
2	7/17	X.guide-cdd	Security guidelines for combining de-identified data using trusted third party	New	Heung Youl Youm, Sungchae Park, Jae Nam Ko	TD978		2021-04	2023-09
3	7/17	X.saf-dfs	Security assurance framework for digital financial services	New	Jacques Francoeur, Jun Hyung Park, Sungchae Park, Heung Youl Youm	TD966		2021-08	2023-09
4	7/17	X.scpa	Security measures for Countering Password Related Online Attacks	New	Hang Dong, Qin Qiu, Lijun Liu, Jung Yeon Hwang, Feng Gao, Jae Hoon Nah	TD1056		2019-08	2023-09
5	10/17	X.pet-auth	Entity authentication service for pet animals using telebiometrics	New	Jae-Sung (Jason) Kim, Taeheon Kim	TD944		2020-03	2023-09
6	10/17	X.osia	Open Standard Identity APIs (OSIA) specification version 6.1.0	New	Abbie Barbir	TD703		2023-03	2023-09
7	11/17	X.508 (X.pki-em)	Public-key infrastructure: Establishment and maintenance	New	Erik Andersen	TD955	ISO/IEC 9594-12	2012-08	2023-09
8	11/17	X.510 Amd.1	Information technology –	New	Erik Andersen	TD954	ISO/IEC 9594-11	2020-08	2023-09

			Open Systems Interconnection – The Directory: Protocol specifications for secure operations						
9	15/17	X.sec-QKDN-tn	Security requirements and designs for quantum key distribution networks - trusted node	New	Qiang Huang, Minghan Li, Jiajun Ma, Hao Qin	TD1021		2019-08	2023-09

d) Non-normative texts (Technical Report, Supplement, Implementers' Guide, etc) planned for agreement

#	Q	Acronym	Title	New / Revised	Editor(s)	Location of text	Start of work	Timing
1.	2/17	TR.cpn-col-sec	Technical Report: Security consideration of collaboration of multiple computing power networks	New	Xiongwei Jia, Zhaoji Lin, Keng Li, Yuwei Wang	TD960	2022-05	2023-09
2.	6/17	TR.ba-iot	Technical Report: Broadcast authentication schemes for IoT system	New	Koji Nakao	TD977	2022-05	2023-09
3.	7/17	TR.sgfdm	Technical Report: FHE-based data collaboration in machine learning	New	Jihoon Cho, Jae Hoon Nah, Donggeon Yhee	TD1035	2020-03	2023-09
4.	14/17	TR.qs-dlt	Technical Report: Guidelines for	New	Fuwen Liu,	TD938	2020-09	2023-09

			quantum-safe DLT system		Ke Wang, Bo Yang, Heung Youl Youm			
5.	15/17	TP.inno-2.0	Technical Paper: Description of the incubation mechanism and ways to improve it	Rev	Arnaud Taddei	TD991	2023.02	2023-09

Annex C

New work items

The following new work items were agreed to be added to the SG17 Work Programme:

#	Question	NWI	Approval	TD	Title	C
1.	1/17	TR.SUSSrev	Agreement	TD1040	Successful use of security standards	
2.	4/17	TR.verm*	Agreement	TD1002	Technical Report: Framework for Verification of Messages	C210
3.	4/17	X.st-ssc*	TAP	TD980	Security threats of software supply chain	C239
4.	6/17	X.mt-integrity*	TAP	TD983	Security guidelines for mobile terminal integrity protection	C273
5.	6/17	X.mt-feature*	TAP	TD984	Security features to assess mobile terminal security	C226
6.	6/17	X.suppl.tig-iotsec**	Agreement	TD950	Supplement to X.1352 (X.suppl.tig-iotsec) "Technical Implementation guidelines for IoT devices and gateway"	C240
7.	7/17	X.srgsc*	TAP	TD942	Security Requirements and guidelines of application and service for smart city platform	C280
8.	8/17	X.asm-cc*	TAP	TD988	Requirements of Attack Surface Management for cloud computing	C287

9.	8/17	X.sfrms*	TAP	TD996	Security framework and requirements of microservice for cloud computing using container technology	C274
10.	10/17	X.bvm	AAP	TD1014	Requirements for biometric variability management	C253
11.	10/17	X.osia	AAP	TD1036	Open Standards Identity APIs (OSIA) version 6.1.0	TD703
12.	11/17	X.jss	AAP	TD925	JSON Signature Scheme (JSS)	C217
13.	11/17	X.509 Cor.1	AAP	TD1029	Technical Corrigendum to X.509: LDAP schema for attribute certificates	TD874
14.	13/17	X.ota-sec	AAP	TD1010	Implementation and evaluation of security functions to support over-the-air (OTA) update capability in connected vehicles	C250
15.	14/17	X.DLT-ccs-fr	AAP	TD962	Security requirements and framework of cross-chain service for DLT systems	C277
16.	15/17	TR.hyb-qsafe**	Agreement	TD1020	Technical Report: Overview of key management of hybrid approaches for quantum-safe communications	C256
17.	15/17	X.sec_QKD_profr	AAP	TD1019	Framework of quantum key distribution (QKD) protocols in QKD network	C269 , C321
18.	15/17	TP.inno-2.0**	Agreement	TD991	Technical Paper: Description of the incubation mechanism and ways to improve it	C216
19.	15/17	X.dtms	AAP	TD995	Guidelines of using digital twin of network for network security	C303
20.	15/17	X.gcspcc*	TAP	TD964	Guidelines of developing of cybersecurity simulation platform based on cloud computing	C314
21.	15/17	X.SecaaS	AAP	TD970	Security threats to be identified in the domain of security as a service	C299

22.	15/17	X.so-sap	AAP	TD941	Guidelines for security orchestration of service access process	C302
23.	15/17	TR.srsec**	Agreement	TD1007	Technical Report: Security aspects of segment routing IPv6 for the convergence of computing and network for telecommunication operators	C286

Annex D

Discontinued work items

Question	Acronym	Title
8/17	X.sr-cphr	Security requirements for cloud-based platform to support low latency and high reliability application scenarios

Annex E

SG17 meeting statistics

Participants ([TD677R1](#))

	participants	countries	Member States	Sector Members	SG17 Associates	Academia	Invited Experts
Announced	332	58	48				
Final	325	54	48	30	2	4	13

Meeting input and organization

Table of SG17 statistics of this and some past meetings

	2023-03	2022-09	2022-05*
C	119	104	101
LS/i	70	55	72
LS/o	23	20	20
TD	394	342	331

Note * - fully virtual meeting

Contributions: 119 – steady increasing, DDP: 98%.

- APT 100.5 (84%) (= China 46.5 + Korea 39 + Japan 10 + India 3 + Singapore 1 + Malaysia 1)
- Americas 14 (12%) (= US 4)

- AFR 2 (Mali, Ghana)
- EUR 1.5 (= UK 1.5)
- RCC 1 (= Russia 1)
- LAM (0), ARAB (0).

LS: matrix in TD699

- incoming **61** - stable
- Outgoing **24** - stable

TDs: 394 – higher than normal